

Example i ① $d \cos(W_t)$ $\left\{ \begin{array}{l} Y_t = f(t, X_t) \\ dY_t = \partial_t f(t, X_t) dt + \partial_x f(t, X_t) dX_t \\ \quad + \frac{1}{2} \partial_{xx} f(t, X_t) (dX_t)^2 \end{array} \right.$

$f(t, x) = \cos x$
 $\partial_t f = 0$, $\partial_x f = -\sin x$, $\partial_{xx} f = -\cos x$

$$d \cos W_t = 0 - \sin(W_t) dW_t - \frac{1}{2} \cos(W_t) dt$$

$(dW_t)^2 = dt$

— x —

② The price S_t of a risky asset evolves according to SDE

$$dS_t = 0.2 S_t dt + S_t dW_t, \quad t \geq 0, \quad S_0 = 5$$

(i) It is suspected that SDE has solⁿ of the form $S_t = c e^{at + bW_t}$, a, b, c const

Use Itô's formula to verify this suspicion and find the const a, b, c

Sol $S_t = c e^{at + bW_t}$

$$f(t, x) = c e^{at + bx}$$

$$\partial_t f = ac e^{at + bx}$$

$$\partial_x f = bc e^{at + bx}, \quad \partial_{xx} f = b^2 c e^{at + bx}$$

$$dS_t = df(t, W_t)$$

$$= ac e^{at + bW_t} dt + bc e^{at + bW_t} dW_t$$

$$+ \frac{1}{2} b^2 c e^{at+bw_t} \underbrace{(dw_t)^2}_{\rightarrow dt}$$

$$= a S_t dt + b S_t dw_t + \frac{1}{2} b^2 S_t dt$$

$$= \left(a + \frac{b^2}{2} \right) S_t dt + b S_t dw_t$$

$$W_0 = 0$$

$$S_0 = 5$$

$$c e^0 = 5 \Rightarrow c = 5$$

$$a + \frac{b^2}{2} = 0.2 \Rightarrow a = -0.3$$

$$b = 1$$

$$b = 1$$

$$c = 5$$

$$S_t = 5 e^{-0.3t + w_t}$$

(ii) Show that the process $X_t = \frac{1}{S_t}$ satisfies the SDE

$$dX_t = 0.3 X_t dt - X_t dw_t, \quad t \geq 0, \quad X_0 = 0.2$$

$$X_t = \frac{1}{S_t} = \frac{1}{5 e^{-0.3t + w_t}} = 0.2 e^{0.3t - w_t} = f(t, w_t)$$

$$f(t, x) = 0.2 e^{0.3t - x}$$

$$\partial_t f = 0.3 \times 0.2 e^{0.3t - x} = 0.3 f$$

$$\partial_x f = -0.2 e^{0.3t - x} = -f$$

$$\partial_{xx} f = 0.2 e^{0.3t-x} = f$$

$$dX_t = df(t, X_t)$$

$$= 0.3 f(t, w_t) dt - f(t, w_t) dw_t + \frac{1}{2} f''(t, w_t) (dw_t)^2$$

$$= 0.3 X_t dt - X_t dw_t + \frac{1}{2} X_t dt$$

$$= 0.8 X_t dt - X_t dw_t$$

Example The "stochastic volatility" Heston model assumes that the "variance process" $(V_t)_{t \geq 0}$ follows the SDE

$$dV_t = (1 - V_t) dt + 2\sqrt{V_t} dw_t, t \geq 0, V_0 = 1$$

(i) Derive an SDE for the "volatility process"

$$Z_t = \sqrt{V_t}, t \geq 0 \text{ and find the initial condition for the SDE (i.e. the value of } Z_0)$$

Sol

$$dZ_t = d\sqrt{V_t} = df(t, V_t)$$

$$f(t, x) = \sqrt{x}$$

$$\partial_t f = 0, \partial_x f = \frac{1}{2} x^{-1/2}, \partial_{xx} f = -\frac{1}{4} x^{-3/2}, x \geq 0$$

$$dZ_t = df(t, V_t)$$

$$= \frac{1}{2} V_t^{-1/2} dV_t - \frac{1}{8} V_t^{-3/2} (dV_t)^2$$

$$= \frac{1}{2} V_t^{-1/2} ((1-V_t)dt + 2\sqrt{V_t} dW_t) - \frac{1}{8} V_t^{-3/2} \times 4V_t dt$$

$$= dW_t + \cancel{\frac{1}{2} V_t^{-1/2} dt} - \frac{1}{2} V_t^{1/2} dt - \cancel{\frac{1}{2} V_t^{-1/2} dt}$$

$$= dW_t - \frac{1}{2} Z_t dt$$

$$dZ_t = dW_t - \frac{1}{2} Z_t dt$$

$$Z_0 = \sqrt{V_0} = 1$$

(ii) Show that the S.P.

$$Z_t := e^{-t/2} \left(1 + \int_0^t e^{s/2} dW_s \right)$$

satisfies the SDE and the initial condition
you derived in part (i),

Sol $Z_t = X_t Y_t$,
where $X_t = e^{-t/2}$

$$Y_t = 1 + \int_0^t e^{s/2} dW_s$$

$$dX_t = -\frac{1}{2} e^{-t/2} dt = -\frac{X_t}{2} dt$$

$$dY_t = e^{t/2} dW_t$$

$$dZ_t = d(X_t Y_t)$$

$$= X_t dY_t + Y_t dX_t + dX_t dY_t$$

$$-t/2 \quad t/2$$

$$= e^{t/2} e^{-t/2} dw_t - \gamma_t \frac{X_t}{2} dt - \frac{X_t}{2} e^{t/2} \underbrace{dt dw_t}_{=0}$$

$$= dw_t - \frac{1}{2} Z_t dt$$